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HIGHER-ORDER QUANTUM CORRECTIONS IN THE PRESENCE OF AN INSTANTON BACKGROUND FIELD

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We calculate the ultraviolet divergent part of the two-loop correction to the vacuum energy in the presence of an instanton background field. We verify that the coupling constant and fermion mass, which appear in the semi-classical calculation of the vacuum energy, indeed become "running" parameters. Among the delicate features of this calculation is the treatment of the zero modes, and in particular the calculation of the corrections to the jacobian corresponding to the change of variables from the zero modes to the collective coordinates.

1. Introduction

Instantons, topologically non-trivial solutions of the euclidean equations of motion in non-abelian gauge theories, were discovered in the mid 1970's [1, 2]. Although their significance may not be fully understood yet, it is clear that they constitute one of the most important non-perturbative effects in gauge theories (for reviews of instanton physics see refs. [3–5]). In this paper we study the problem of how to calculate quantum corrections to the semi-classical approximation; in particular we calculate the two-loop contribution to the vacuum energy in the presence of an instanton background field. Although the renormalisation group tells us what the ultraviolet divergences must be (and hence what the dependence of the renormalized quantities on the subtraction point must be) it has proved instructive to see how the divergences from three different sources combine to give the expected result. These three sources are

(i) divergences from the purely short-distance region of phase space (these can be calculated perturbatively),

(ii) divergences from regions of phase space in which some long-distance effects are important (these have to be studied non-perturbatively),

(iii) divergences which arise when calculating higher-order terms in the jacobian which is present when one changes the integration variables from the zero modes to the collective coordinates.

In this paper we restrict ourselves to the specific case of a single instanton background field, however it is clear that the same methods apply to calculations in the presence of any background field which is a solution of the equations of motion and which breaks some of the symmetries of the lagrangian.

In a recent paper Novikov et al. [6] have considered the vacuum energy in an instanton background field in supersymmetric Yang-Mills theories. Using formal arguments they conclude that there are no corrections to the semi-classical answer, i.e. the instanton contribution to the renormalised vacuum energy is proportional to

$$\int \mathrm{d}^4 x_0 \frac{\mathrm{d}\rho}{\rho^5} \mathrm{d}^2\alpha \, \mathrm{d}^2\bar{\beta} \frac{1}{[g(\mu)]^{n_\mathrm{b}-n_\mathrm{f}}} (\mu\rho)^{n_\mathrm{b}-n_\mathrm{f}/2} \mathrm{e}^{-8\pi^2/g^2(\mu)}, \tag{1.1}$$

where the collective coordinates x_0 and ρ are the "centre" and "size" of the instanton, and the two Grassmann collective coordinates α and $\bar{\beta}$ correspond to supersymmetry and superconformal transformations respectively. $g(\mu)$ is the renormalised coupling constant, μ being the renormalisation subtraction point and n_b and n_f are the numbers of bosonic and fermionic zero modes (the fermions are assumed to be Majorana fermions). From the remarkable result that there are no higher-order corrections to (1.1), and noting that for a $n = 1$, 2 or 4 supersymmetric SU(N) theory $n_\mathrm{b} = 4N$ and $n_\mathrm{f} = 2Nn$, one concludes that

$$\beta(g) \equiv \frac{\mathrm{d}g(\mu)}{\mathrm{d}\ln\mu} = \frac{g^3(\mu)}{16\pi^2} \frac{(n-4)N}{1-((2-n)/8\pi^2)Ng^2(\mu)}, \tag{1.2}$$

in agreement, at least for the first two terms in the series in g^2, with standard results. The same β-function has been obtained by considering the multiplet of anomalies in supersymmetric theories [7]. Ref. [6] raises many interesting questions in supersymmetric theories which we hope to discuss elsewhere. In this paper we restrict our attention to ordinary Yang-Mills theories, where of course there are higher-order corrections on the right-hand side of the analogous equation to (1.1), however the $\ln\mu$ dependence of these higher-order terms is determined by the renormalisation group. This provides a useful check on calculations in an instanton background field.

Before going on to the details of the calculation in the instanton background field, we review briefly the "background field method" in which one calculates quantities in the presence of an unspecified background field (for a review and references to the original literature see [8]). The method is based on the background field gauge invariance of the lagrangian density.

$$\mathcal{L} = \tfrac{1}{4}\tilde{F}_{\mu\nu}\tilde{F}_{\mu\nu} - i\bar{\psi}(\gamma_\mu \tilde{D}_\mu + m)\psi. \tag{1.3}$$

We write the lagrangian density in euclidean space in order to make closer contact with the instanton calculation below, and introduce the tilde over the field strength tensor $\tilde{F}_{\mu\nu}$ and covariant derivative $\tilde{D}_\mu$ to distinguish these quantities from the corresponding ones which involve the classical background field only, which we denote as $F_{\mu\nu}$ and D_μ respectively. In euclidean space,

$$\tilde{F}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig\left[A_\mu, A_\nu\right], \tag{1.4}$$

$$\tilde{D}_\mu = \partial_\mu + igA_\mu, \tag{1.5}$$

with $A_\mu \equiv A_\mu^a \tau^a$ and $\tilde{F}_{\mu\nu} \equiv \tilde{F}_{\mu\nu}^a \tau^a$ where the τ^a are the generators of the gauge group (henceforth we suppress the group indices and $\tilde{D}_\mu$, A_μ and $\tilde{F}_{\mu\nu}$ are understood to be matrices in group space). We now expand around a classical background field A_μ^{cl}:

$$A_\mu = A_\mu^{\mathrm{cl}} + Q_\mu, \tag{1.6}$$

and a convenient choice for the gauge-fixing term is

$$\mathcal{L}_{\mathrm{GF}} = \frac{1}{2\xi}\left(D_\mu Q_\mu\right)^2. \tag{1.7}$$

With this choice the theory can be seen to be invariant under the background field gauge transformation

$$A_\mu^{\mathrm{cl}} \to \mathrm{e}^{-i\Omega} A_\mu^{\mathrm{cl}} \mathrm{e}^{i\Omega} - \frac{i}{g}\mathrm{e}^{-i\Omega}\,\partial_\mu\,\mathrm{e}^{i\Omega}, \tag{1.8}$$

as can be seen by simultaneously making a homogeneous change of variables (with jacobian equal to 1) in the functional integral over the quantum field;

$$Q_\mu \to \mathrm{e}^{-i\Omega} Q_\mu \mathrm{e}^{i\Omega}. \tag{1.9}$$

(1.8) together with (1.9) constitute the usual gauge transformation on the full field A_μ;

$$A_\mu \to \mathrm{e}^{-i\Omega} A_\mu \mathrm{e}^{i\Omega} - \frac{i}{g}\mathrm{e}^{-i\Omega}\,\partial_\mu\,\mathrm{e}^{i\Omega}. \tag{1.10}$$

The invariance under background field gauge transformations (1.8) implies that all the infinities in the higher-order corrections to the effective action are proportional to $F_{\mu\nu}F_{\mu\nu}$ and that $gF_{\mu\nu}$ remains unrenormalised. Thus the corresponding renormalised quantity to (1.1) in a Yang-Mills theory in the presence of an arbitrary background field A_μ^{cl} ($\langle 0|0\rangle_{A^{\mathrm{cl}}}/\langle 0|0\rangle$) is proportional to

$$\exp\left(-\tfrac{1}{4}\int F_{\mu\nu}F_{\mu\nu}\left(1-\frac{2\beta_0}{16\pi^2}g^2(\mu)\ln\mu-\frac{2\beta_1}{(16\pi^2)^2}g^4(\mu)\ln\mu\ldots\right)\mathrm{d}^4x\right), \quad (1.11)$$

where our notation is

$$\beta(g)\equiv\frac{\partial g(\mu)}{\partial\ln\mu}=-\frac{\beta_0}{16\pi^2}g^3(\mu)-\frac{\beta_1}{(16\pi^2)^2}g^5(\mu)+\cdots. \quad (1.12)$$

All the $\ln\mu$ dependence in the higher-order terms is determined by the β-function. Thus calculating the higher-order corrections to the effective action is a method, and as it turns out a useful method, of determining the β-function.

We wish to stress that the ultraviolet infinities (and hence the $\ln\mu$ dependence) come entirely from the short-distance region of phase space. To illustrate this let us consider the two-loop diagrams of fig. 1. Fig. 1a has a vertex insertion and therefore a corresponding ultraviolet divergence coming from a large momentum in one loop, even if the momentum in the outer loop is small. However since gA_μ^{cl} is unrenormalised, there is a $Z_1=Z_2$ Ward identity (Z_1 refers to the renormalisation of the coupling of the background field to the quantum field) and this divergence is cancelled by the diagram of fig. 1b with a self-energy insertion. Moreover there are clearly the same number of two-loop diagrams with a vertex insertion infinity as there are those with a self-energy infinity. Thus the only remaining infinities are from overlapping divergences, that is those from the region of phase space in which the momenta in both loops are large simultaneously. This can be generalised to higher-order terms in the loop expansion, although, of course, beyond two loops some of the ultraviolet divergences will be absorbed into coupling constant renormalisation.

Since the $\ln\mu$ dependence in (1.11) comes from the short-distance region of phase space, it is independent of the topological properties of the background field and is

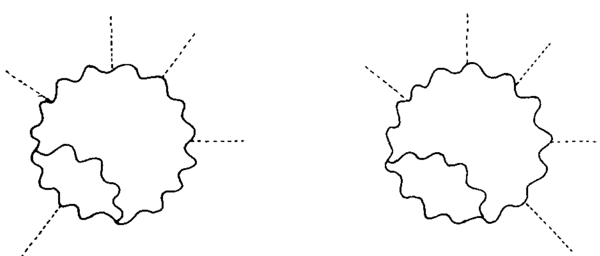

Fig. 1. Two-loop diagram with ultraviolet divergence from vertex insertion. (b) Corresponding two-loop diagram with ultraviolet divergence from self-energy insertion.

also present in the case of an instanton background field. In the instanton case however there are additional contributions, a determination of these requires a detailed knowledge of the number and properties of the zero modes. In sect. 2 we discuss these zero modes in detail, although for convenience we postpone the consideration of fermions until Sect. 5. Since we will be interested in going beyond the semi-classical approximation we need a systematic way of evaluating the jacobian corresponding to the change of variable from the zero modes to the collective coordinates. In analogy to the Faddeev-Popov procedure we evaluate this jacobian by introducing a Grassmann variable for each zero mode, and deriving the corresponding effective lagrangian and Feynman rules. These Grassmann variables are x-independent since the corresponding zero modes are discrete. The effective lagrangian contains new terms of the type $\bar{c}Q\phi + \bar{\phi}Qc$ where c is the new Grassmann variable, ϕ is the Faddeev-Popov ghost and Q is the quantum gauge field. The presence of these terms is a manifestation of the fact that the gauge-fixing procedure and the elimination of the discrete zero modes are coupled problems, and must be treated simultaneously.

In sect. 3 we review the semi-classical evaluation of the vacuum energy in the presence of an instanton background field, carefully exhibiting all the $\ln\mu$-dependent quantities. Sect. 4 contains the two-loop calculations for the pure gauge theory. We verify that the $\ln\mu$ dependence is as expected from the results of sect. 3 in the semi-classical approximation. Among these $\ln\mu$ terms are those arising from extra one-loop diagrams generated by the correction to the jacobian which we evaluate using the Feynman rules determined in sect. 2.

In sect. 5 we discuss the inclusion of fermions. For strictly massless fermions there are fermionic zero modes which render the instanton contribution to the vacuum energy zero. We deal with this by giving the fermions a small mass. The contribution from the semi-classical approximation includes a factor of the mass, m, for each fermionic zero mode. In higher orders we show that there are extra $\ln\mu$ contributions which come from the integration over the zero modes of the massless theory, which compensate for the μ-dependence of the mass parameter, m.

Sect. 6 contains a summary and our conclusions.

2. Zero modes

The leading-order (semi-classical) approximation to the vacuum-vacuum amplitude in the presence of a classical background field is obtained by expanding the full field into the classical background field plus the quantum part and neglecting cubic and higher-order terms in the quantum field. For Yang-Mills (vector) fields this means

$$A_\mu(x) = A_\mu^{\mathrm{cl}}(x) + Q_\mu(x)\,. \tag{2.1}$$

Expanding the action up to quadratic terms in the quantum field we obtain

$$S = S_{\mathrm{cl}} + \int \mathrm{d}^4x \left[f_\mu(x) Q_\mu(x) + \tfrac{1}{2} Q_\mu(x) O_{\mu\nu} Q_\nu(x) \right], \tag{2.2}$$

so that in euclidean space we have

$$\langle 0|0\rangle_{A_{\mathrm{cl}}} = \int D[Q_\mu(x)] \exp\left[-\left\{ S_{\mathrm{cl}} + \int \mathrm{d}^4x \left[f_\mu(x) Q_\mu(x) + \tfrac{1}{2} Q_\mu O_{\mu\nu} Q_\nu(x) \right] \right\} \right]$$

$$\sim \frac{\exp\left[-\left\{ S_{\mathrm{cl}} - \int \mathrm{d}^4x\, \mathrm{d}^4y\, \tfrac{1}{2} f_\mu(x) O_{\mu\nu}^{-1}(x, y) f_\nu(y) \right\} \right]}{\Pi \sqrt{E_i}}, \tag{2.3}$$

where E_i are the eigenvalues of the operator $O_{\mu\nu}$ which for a Yang-Mills theory is given by

$$O_{\mu\nu} = -D^2 \delta_{\mu\nu} + D_\mu D_\nu - 2igF_{\mu\nu}, \tag{2.4}$$

where $D_\mu = \partial_\mu + igA_\mu^{\mathrm{cl}}$ and $igF_{\mu\nu} = [D_\mu, D_\nu]$.

For an instanton, which is a solution of the (euclidean) equations of motion the linear term $(f_\mu Q_\mu)$ vanishes, however this formalism cannot be used since there are several eigenvectors of $O_{\mu\nu}$ with zero eigenvalues. These zero modes correspond to the generators of the symmetry of the action which are broken by the instanton solution. This is because there is an infinitely degenerate number of instanton solutions and the zero modes correspond to transformation between these degenerate solutions.

Since the instanton solution

$$A_\mu^I(x) = \frac{2}{g} \frac{\eta_{\mu\nu}(x^\nu - a^\nu)}{(x-a)^2 + \rho^2} \tag{2.5}$$

($\eta_{\mu\nu} = \tau^a \eta_{\mu\nu}^a$ where τ^a are the generators of an SU(2) subgroup of the gauge group (SU(N)) and $\eta_{\mu\nu}^a$ are the 't Hooft symbols [1,2]) has a centre at $x = a$ and a size ρ it breaks both translational invariance and dilatation invariance so that there are five zero modes associated with translations (four) and dilatations. Furthermore the global SU(N) group and the rotational O(4) group are broken down to SU($N-2$) $\times$ U(1) $\times$ O′(4) where the generators of O′(4) are a linear combination of the generators of O(4) and some of the generators of SU(N) which are not contained in the unbroken subgroup SU($N-2$) $\times$ U(1). This leads to a further $4N-5$ zero modes*. In order to be able to perform calculations in an instanton background

* For the explicit form of these zero modes see ref. [2].

field these zero modes must be eliminated and replaced in the path integral by integrations over collective coordinates corresponding to the position, size, and group rotation of the instanton.

The instanton also breaks the local gauge invariance of the group, i.e. the gauge rotated instanton

$$e^{-i\Omega(x)}A_\mu(x)e^{i\Omega(x)}-\frac{i}{g}e^{-i\Omega(x)}\,\partial_\mu\,e^{i\Omega(x)} \tag{2.6}$$

is also a solution so we have an infinite number of zero modes corresponding to the infinite set of possible gauge functions $\Omega(x)$.

The zero mode associated with each $\Omega(x)$ is given by $D_\mu\Omega(x)$. The procedure of fixing a gauge is just the elimination of these zero modes and their replacement in the path integral by integration over the gauge group. The Faddeev-Popov part of the effective action is the jacobian arising from this change of variables. However as has been pointed out in ref. [9] (see also ref. [10]) since the discrete zero modes are gauge dependent and the gauge zero modes depend on the centre, size, etc. of the instanton the procedure of eliminating all the zero modes must be performed simultaneously.

One usually distinguishes the gauge transformations $\Omega(x)$ from the global group transformations by demanding $\Omega(x)\to 0$, as $|x|\to\infty$. The functions $\Omega(x)$ can conveniently be expanded:

$$\Omega(x)=\int d\alpha\, b(\alpha)\Omega(\alpha,x), \tag{2.7}$$

where $\Omega(\alpha, x)$ is an eigenstate of $-D^2$:

$$-D^2\Omega(\alpha,x)=\lambda^2(\alpha)\Omega(\alpha,x),\quad \Omega(\alpha,x)\to 0\quad \text{as } |x|\to\infty .$$

If we relax this boundary condition and allow $\Omega(x)$ to remain finite but not necessarily zero as $|x|\to\infty$, the global transformations $\Omega(x)=$ constant are included. However as has been shown in ref. [2] we can now separate these functions Ω into two sets. The first set contains those Ω for which

$$x^\mu\Omega(x)\,\partial_\mu\Omega(x)\underset{|x|\to\infty}{\to} 0\quad \text{faster than}\quad 1/|x|^2,$$

for which a quantity like $(D_\mu\Omega)^2$ may be integrated by parts as is required in the usual Faddeev-Popov formalism (note that the global transformations $\Omega(x)=$ constant fall into this class). This implies that $-\int d^4x\,\Omega D^2\Omega$ is positive definite (apart from Ω corresponding to global transformations in the unbroken subgroup, which are irrelevant since they do not give zero modes) and so these functions may be expanded in terms of the positive eigenmodes of $-D^2$. It is the zero modes of these gauge transformations which are eliminated by the imposition of a background field gauge condition. The second set consists of further functions $\Omega^a(x)$ ($a = 1,\dots,4N-5$) for which $x^\mu \Omega^a(x)\,\partial_\mu \Omega^a(x)$ vanishes more slowly than $1/|x|^2$ as $|x| \to \infty$ so that $|D_\mu \Omega^a|^2$ may not be integrated by parts. These give $4N-5$ further zero modes even after the imposition of the gauge condition. It can therefore be shown that for these functions, $-D^2\Omega^a(x) = 0$, which is possible because the boundary condition that Ω vanishes as $|x| \to \infty$, has been lifted [2]. Thus the total gauge transformation (including global transformations) may be parameterized by

$$\Omega(\theta^a, b(\alpha), x) = \sum_{a=1}^{4N-5} \theta^a \Omega^a(x) + \int \mathrm{d}\alpha\, b(\alpha)\Omega(\alpha, x), \tag{2.8}$$

where

$$-D^2\Omega^a(x) = 0,$$

$$-D^2\Omega(\alpha, x) = \lambda^2(\alpha)\Omega(\alpha, x) \qquad (\text{with } \Omega^a(x), \Omega(\alpha, x) \text{ finite as } |x| \to \infty).$$

We can now write the most general translated gauge and global transformed instanton of size $\tilde{\rho}$, as

$$A^I_\mu(b^i, b(\alpha), x) = \mathrm{e}^{-i\Omega(\theta^a, b(\alpha), x)} \frac{2}{g} \frac{\eta_{\mu\nu}(x^\nu - a^\nu)}{(x-a)^2 + \tilde{\rho}^2} \mathrm{e}^{+i\Omega(\theta^a, b(\alpha), x)}$$

$$- \frac{i}{g} \mathrm{e}^{-i\Omega(\theta^a, b(\alpha), x)} \partial_\mu \mathrm{e}^{+i\Omega(\theta^a, b(\alpha), x)}, \tag{2.9}$$

where b^i stands for the $4N$ parameters $a_\mu, \tilde{\rho} - \rho, \rho\theta^a$ (ρ is the arbitrarily chosen instanton size when all b^i are set to zero, and the parameters θ^a have been multiplied by the instanton size ρ so that all the parameters b^i have the same dimension M^{-1}). The zero modes are now given by

$$\zeta^{[p]}_\mu(b^i, b(\alpha), x) = g \frac{\partial}{\partial b^{[p]}} A^I_\mu(b^i, b(\alpha), x), \tag{2.10}$$

where $[p]$ runs over the discrete indices ($= 1 \dots 4N$) and the continuous variable α, in particular the gauge zero modes are

$$\zeta^\beta_\mu(b^i, b(\alpha), x) = g \frac{\delta}{\delta b(\beta)} A^I_\mu(b^i, b(\alpha), x), \tag{2.11a}$$

where β is a continuous variable. Taking linear combinations these zero modes may be written

$$\zeta^\beta_\mu(b^i, b(\alpha), x) = D_\mu \Omega(\beta, x), \tag{2.11b}$$

and the discrete zero modes are given by

$$\zeta_\mu^i(b^j, b(\alpha), x) = g\frac{\partial}{\partial b^i}A_\mu^I(b^j, b(\alpha), x). \tag{2.12}$$

In order to have orthogonal zero modes it is necessary to add a linear sum of the gauge zero modes to some of the discrete zero modes [2, 9]. Alternatively this may be expressed by keeping eq. (2.12) unchanged but redefining some of the b^i by adding a linear sum of $b(\alpha)$ to them. The factor g is introduced into the definition of the zero modes so that the modes have coupling constant independent norms (A_μ^I is proportional to $1/g$).

The zero modes have an explicit dependence on the b (both the discrete set b^i and the continuous function $b(\alpha)$):

$$\zeta_\mu^{[p]}(b, x) = \frac{\tilde{\rho}^2}{\rho^2}\mathrm{e}^{-i\Omega(\theta^a, b(\alpha), x)}\zeta_\mu^{[p]}\left(0, \frac{\rho}{\tilde{\rho}}(x-a)\right)\mathrm{e}^{+i\Omega(\theta^a, b(\alpha), x)}. \tag{2.13}$$

The full field is now expanded as

$$A_\mu(x) = A_\mu^I(b, x) + Q_\mu(x). \tag{2.14}$$

Note that under a gauge transformation $Q_\mu(x)$ transforms homogeneously and it is the classical part that carries the inhomogeneous part.

The identity is introduced into the path integral in the following way:

$$\begin{aligned} 1 = \int & D[b(\alpha)]\,\mathrm{d}^{4N}b^i\,D[K(\alpha)]\,\mathrm{e}^{-(1/2\xi)\int K^2(\alpha)\,\mathrm{d}\alpha} \\ & \times \prod_\alpha \delta\left(\int \mathrm{d}^4x\left[Q_\mu(x)D_\mu\Omega(\alpha, x)\right] - K(\alpha)\right) \\ & \times \frac{\prod_i \delta\left[\int \mathrm{d}^4x\,Q_\mu(x)\zeta_\mu^i(b, x)\right]\det J}{\int D[K(\alpha)]\exp\left[-(1/2\xi)\int K^2(\alpha)\,\mathrm{d}\alpha\right]}. \end{aligned} \tag{2.15}$$

The integration over $K(\alpha)$ gives the usual gauge-fixing term:

$$-\frac{1}{2\xi}\int \mathrm{d}^4x\,(D_\mu Q_\mu)^2,$$

in the effective action.

The remaining delta functions eliminate the discrete zero modes, by imposing the condition that the quantum fluctuations Q_μ are orthogonal to all those zero modes. The jacobian term $\det J$ is the determinant of the extended matrix:

$$J^{[p],[q]} = \frac{\partial}{\partial b^{[q]}} \int \mathrm{d}^4x\, Q_\mu(x) \zeta_\mu^{[p]}(b, x), \tag{2.16}$$

where $[p]$ and $[q]$ both run over the indices $i = 1 \dots 4N$ and the variable α.

Now the full field $A_\mu(x)$ cannot depend on the parameters b of the background field about which one chooses to expand. Therefore using eq. (2.14) we have

$$\frac{\partial}{\partial b^{[q]}} Q_\mu(x) = -\frac{\partial}{\partial b^{[q]}} A_\mu^I(b, x) = -\frac{1}{g} \zeta_\mu^{[q]}(b, x). \tag{2.17}$$

Thus (2.16) may be written

$$J^{[p],[q]} = -\frac{1}{g} \int \mathrm{d}^4x\, \zeta_\mu^{[p]}(b, x) \zeta_\mu^{[q]}(b, x)$$
$$+ \int \mathrm{d}^4x \left(Q_\mu(x) \frac{\partial}{\partial b^{[q]}} \zeta_\mu^{[p]}(b, x) \right), \tag{2.18}$$

where the orthogonality of the zero modes means that the first term on the right-hand side of eq. (2.18) is diagonal. The determinant is obtained by introducing the infinite set of Grassmann variables $c^{[p]}$ $(= c(\alpha), c^i)$ and their conjugates, $\bar{c}(\alpha), \bar{c}^i$ and adding the term $\bar{c}^{[q]} J^{[p],[q]} c^{[p]}$ to the effective action. To leading order we have

$$\det J = \int D[\bar{c}^{[r]}] D[c^{[r]}] \exp\left[-\bar{c}^{[p]} \frac{1}{g} \int \mathrm{d}^4x\, \zeta_\mu^{[p]} \zeta_\mu^{[q]} c^{[q]} \right]. \tag{2.19}$$

We define

$$\frac{1}{\sqrt{g}} \int \mathrm{d}\alpha\, c(\alpha) \Omega(\alpha)$$

to be the usual Faddeev-Popov scalar $\phi(x)$ and

$$\frac{1}{\sqrt{g}} \int \mathrm{d}\alpha\, \bar{c}(\alpha) \Omega(\alpha, x)$$

to be $\bar{\phi}(x)$, so that to this order we have

$$\det J = \int \prod \mathrm{d}\bar{c}^i \mathrm{d}c^i \exp\left[-\frac{1}{g} \bar{c}^i c^i \int \mathrm{d}^4x\, \zeta^i_\mu \zeta^i_\mu \right] D[\bar{\phi}(x)]\, D[\phi(x)]$$

$$\times \exp\left[\int \mathrm{d}^4x\, \bar{\phi} D^2 \phi \right], \tag{2.20}$$

which gives us the propagator for the usual Faddeev-Popov ghost term in the background field gauge and a factor $-(1/g)N_i^2$, where N_i is the norm of the ith discrete zero mode, for each discrete zero mode.

If we wish to go beyond the semi-classical approximations we must also consider the second term of eq. (2.18). We consider separately the cases in which $[p]$, $[q]$ = continuous and discrete.

(i) $[p]$ and $[q]$ both continuous, $[p] \equiv \alpha$, $[q] \equiv \beta$:

$$\frac{\delta}{\delta b(\beta)} \zeta^\alpha_\mu(b, x) = \frac{\delta}{\delta b(\beta)} D_\mu \Omega(a, x)$$

$$= -i\left[\Omega(\beta, x), D_\mu \Omega(\alpha, x) \right]. \tag{2.21}$$

(This is the change in $D_\mu \Omega(\alpha, x)$ under an infinitesimal gauge transformation $\Omega(\beta, x)$.) Thus the contribution to the effective action is

$$\int \mathrm{d}\alpha\, \mathrm{d}\beta\, \mathrm{d}^4x\, \bar{c}(\beta) Q_\mu(x) \left[\Omega(\beta, x), D_\mu \Omega(\alpha, x) \right] c(\alpha)$$

$$= -ig \int \mathrm{d}^4x\, Q_\mu(x) \left[\bar{\phi}(x), D_\mu \phi(x) \right]. \tag{2.22}$$

This is the usual expression for the vertex between the gauge field and the Faddeev-Popov ghost field (in the background field gauge).

(ii) $[p]$ and $[q]$ both discrete $[p] \equiv i$, $[q] \equiv j$. This gives rise to a term in the effective action which is linear in the quantum field Q_μ:

$$\int \mathrm{d}^4x\, \bar{c}^j Q_\mu(x) \frac{\partial}{db^j} \zeta^i_\mu(b, x) c^i. \tag{2.23}$$

Such interactions can only occur on external lines and not inside loops. They therefore only give rise to finite corrections to the vacuum-vacuum amplitude.

(iii) $[p]$ continuous, $[q]$ discrete, $[p]\equiv\beta$, $[q]=j$.

$$\zeta_\mu^{[p]}=D_\mu\Omega(\beta,x)=\partial_\mu\Omega(\beta,x)+ig\left[A_\mu^I(b,x),\Omega(\beta,x)\right], \qquad (2.24)$$

$$\frac{\partial\zeta_\mu^\beta(b,x)}{\partial b^j}=i\left[\zeta_\mu^j(b,x),\Omega(\beta,x)\right]+D_\mu\frac{\partial}{\partial b^j}\Omega(\beta,x). \qquad (2.25)$$

The functions Ω are implicit functions of b^i since they are eigenstates of D^2 where $D_\mu=\partial_\mu+igA_\mu^I(b^i,b(\alpha),x)$ and so the contribution of this to the effective action is

$$i\bar{c}^j\int d\beta\, d^4x\, Q_\mu(x)\left[\zeta_\mu^j(b,x),\Omega(\beta,x)\right]c(\beta)$$

$$+i\bar{c}^j\int d\beta\, d^4x\, Q_\mu(x)D_\mu\frac{\partial}{\partial b^j}\Omega(\beta,x)c(\beta). \qquad (2.26)$$

The second term after integration by parts gives a bilinear interaction proportional to $D_\mu Q_\mu$ which by the Slavnov-Taylor identities [11] in the background field gauge cannot contribute to physical quantities (such as vacuum energy). We are thus left with

$$i\sqrt{g}\,\bar{c}^i\int d^4x\, Q_\mu(x)\left[\zeta_\mu^i,\phi\right]. \qquad (2.27)$$

This represents a bilinear coupling between the gauge field and the Faddeev-Popov ghost fields*. As will be shown in sect. 4 such a coupling gives rise to important extra loop contributions, when working beyond the (leading-order) semi-classical approximation.

(iv) $[p]$ discrete, $[q]$ continuous, $[p]\equiv i$, $q\equiv\alpha$:

$$\frac{\delta}{\delta b(\alpha)}\zeta_\mu^i(b,x)=-i\left[\Omega(\alpha,x),\zeta_\mu^i(b,x)\right]. \qquad (2.28)$$

Again this is the change in the mode ζ_μ^i under an infinitesimal gauge transformation

* A derivation of this coupling using the explicit form of the zero modes can be found in ref. [9], for the zero modes corresponding to translations and dilatations.

Fig. 2. Bilinear couplings arising from the jacobian. (a) $\sqrt{g}\,\bar{c}^i\zeta^i_{\mu,c}(x)f^{abc}$, (b) $\sqrt{g}\,c^i\zeta^i_{\mu,c}(x)f^{abc}$.

$\Omega(\alpha, x)$ and this contributes to the effective action

$$-i\int \mathrm{d}\alpha\,\mathrm{d}^4x\,\bar{c}(\alpha)Q_\mu(x)\left[\Omega(\alpha, x), \zeta^i_\mu(b, x)\right]c^i$$

$$= -i\sqrt{g}\int \mathrm{d}^4x\,Q_\mu(x)\left[\bar{\phi}(x), \zeta^i_\mu(b, x)\right]c^i. \tag{2.29}$$

This represents a bilinear coupling between the gauge field and the conjugate Faddeev-Popov field and again this gives rise to important extra loop contributions beyond the semi-classical approximation.

The Feynman rules for both these bilinear couplings are shown in fig. 2.

Finally we point out that having introduced the parameters b^i and $b(\alpha)$ in order to define the zero modes and the determinant of the matrix $J^{[p],[q]}$, we can for convenience then make the relevant gauge and coordinate transformations back to $b^i \equiv b(\alpha) \equiv 0$, i.e. the extra linear and bilinear coupling given by eqs. (2.23), (2.27) and (2.29) may be calculated using $\zeta^i_\mu(0, x)$.

Let us briefly summarize the results of this section. We have performed the elimination of all the zero modes associated with an instanton background field, and have shown that the gauge-fixing procedure and the treatment of zero modes must be performed simultaneously. In addition to the usual Faddeev-Popov term in the effective action we find extra terms (2.23), (2.27) and (2.29). In the following sections we shall need the Feynman rules for the latter two terms, and these are given in fig. 2.

3. The semi-classical approximation

From the previous section we see that the leading-order contribution to the ratio of the vacuum-vacuum amplitude in the presence of an instanton to the vacuum-

vacuum amplitude in the zero instanton sector may be written as

$$\frac{\langle 0|0\rangle_I}{\langle 0|0\rangle_0}=C\frac{1}{g^{4N}}\frac{\int \mathrm{d}^{4N}b\,\mu^{4N}D'[Q_\mu]D[\bar{\phi}]D[\phi]\times\exp\left(-\frac{8\pi^2}{g^2}-\int \mathrm{d}^4x\,\frac{1}{2}Q_\mu O_{\mu\nu}Q_\nu-(1/2\xi)Q_\mu D_\mu D_\nu Q_\nu-\bar{\phi}D^2\phi\right)}{\int D[Q_\mu]D[\bar{\phi}]D[\phi]\exp\int \mathrm{d}^4x\left(+\frac{1}{2}Q_\mu\partial^2 Q_\mu-\frac{1}{2}(1-1/\xi)Q_\mu\partial_\mu\partial_\nu Q_\nu+\bar{\phi}\partial^2\phi\right)} \tag{3.1}$$

where the operator $O_{\mu\nu}$ is given in eq. (2.4) and $D'[Q_\mu]$ means integrate over the non-zero eigenvalue modes of $O_{\mu\nu}-(1/\xi)D_\mu D_\nu$. The term $8\pi^2/g^2$ in the exponential in the numerator is the classical instanton action. The factors of $1/g$ as well as the term $\bar{\phi}D^2\phi$ in the effective action arise from the leading-order part of det J, the jacobian which arises from the elimination of zero modes and which is discussed in the previous section. C is a numerical constant. The integration over the gauge group $D[b(\alpha)]$ cancels between numerator and denominator, but the integration over the $4N$ collective coordinates associated with the discrete modes in the one-instanton sector must be performed. The expression (3.1) refers to the renormalized quantity and the coupling constant g must be thought of as a function of the renormalisation subtraction point, μ. The presence of the factor μ^{4N} can be seen by dimensional analysis (the collective coordinates b_i have dimension M^{-1}), μ being the only other dimensionful parameter available (an alternative argument for the presence of this factor when Pauli-Villars regularization is used to regulate infinities can be found in refs. [2,4]). The fields ϕ and $\bar{\phi}$ are Grassmann fields. Performing the gaussian integrals over Q_μ, ϕ and $\bar{\phi}$ we obtain

$$\frac{\langle 0|0\rangle_I}{\langle 0|0\rangle_0}$$

$$=C'\frac{1}{g^{4N}}\int\frac{\mathrm{d}^{4N}b_i\,\mathrm{e}^{-(8\pi^2/g^2-4N\ln\mu)}\det(-D^2)\left(\det\left(-\partial^2\delta\mu_\nu+(1-1/\xi)\,\partial_\mu\partial_\nu\right)\right)^{1/2}}{\rho^{4N}\det(-\partial^2)\left(\det'\left(O_{\mu\nu}-(1/\xi)\,D_\mu D_\nu\right)\right)^{1/2}}, \tag{3.2}$$

where det′ means the product over the positive-definite eigenvalues. (We have multiplied and divided by ρ^{4N} so that the integral over collective coordinates appears dimensionless.) It has been shown in refs. [12,13] that each positive frequency mode of the operator $O_{\mu\nu}-(1/\xi)D_\mu D_\nu$ is associated with an eigenmode of $-D^2$. Furthermore for each eigenmode of $-D^2$,

$$-D^2\Omega(\alpha,x)=\lambda^2(\alpha)\Omega(\alpha,x), \tag{3.3}$$

there are four eigenmodes of $O_{\mu\nu}-(1/\xi)D_\mu D_\nu$. Three of these are given by

$$\bar{\eta}^a_{\mu\nu}D_\nu\Omega(\alpha,x), \qquad a=1,\dots 3,$$

where $\bar{\eta}^a_{\mu\nu}$ are the 't Hooft symbols corresponding to an anti-instanton [2], and for these three eigenmodes

$$O_{\mu\rho}\bar{\eta}^a_{\rho\nu}D_\nu\Omega(\alpha,x)=\lambda^2(\alpha)\bar{\eta}^a_{\mu\nu}D_\nu\Omega(\alpha,x), \tag{3.4a}$$

$$D_\mu\bar{\eta}^a_{\mu\nu}D_\nu\Omega(\alpha,x)=0, \tag{3.4b}$$

so these three have eigenvalue $\lambda^2(\alpha)$. The fourth mode is $D_\mu\Omega(\alpha,x)$ for which

$$O_{\mu\nu}D_\nu\Omega(\alpha,x)=0, \tag{3.5a}$$

$$-(1/\xi)D_\mu D_\nu D_\nu\Omega(\alpha,x)=(1/\xi)\lambda^2(\alpha)D_\mu\Omega(\alpha,x). \tag{3.5b}$$

Therefore we have for the product of all the eigenvalues

$$\det{}'\big(O_{\mu\nu}-(1/\xi)D_\mu D_\nu\big)=\prod_\alpha\frac{1}{\xi}\left(\lambda^2(\alpha)\right)^4=\left(\frac{1}{\xi}\right)^{n_{\mathrm{L}}}\left[\det(-D^2)\right]^4, \tag{3.6}$$

where n_{L} is the number of longitudinal positive modes of the operator

$$O_{\mu\nu}-(1/\xi)D_\mu D_\nu.$$

Similarly in the zero-instanton sector, to each eigenmode of $-\partial^2$

$$-\partial^2u(\alpha,x)=k^2(\alpha)u(\alpha,x), \tag{3.7}$$

there correspond four eigenmodes of $-\partial^2\delta_{\mu\nu}+\partial_\mu\partial_\nu-(1/\xi)\partial_\mu\partial_\nu$, namely

$$\left(\delta_{\mu\nu}-\frac{\partial_\mu\partial_\nu}{\partial^2}\right)u(\alpha,x),$$

which is transverse and for which

$$\left(-\partial^2\delta_{\mu\rho}+\partial_\mu\partial_\rho\right)\left(\delta_{\rho\nu}-\frac{\partial_\nu\partial_\rho}{\partial^2}\right)u(\alpha,x)=k^2(\alpha,x)\left(\partial_{\rho\nu}-\frac{\partial_\nu\partial_\rho}{\partial^2}\right)u(\alpha,x) \tag{3.8}$$

and $\partial_\mu u(\alpha, x)$, which is longitudinal and for which

$$-\frac{1}{\xi}\partial_\mu \partial_\nu \partial_\nu u(\alpha, x) = \frac{k^2(\alpha)}{\xi}\partial_\mu u(\alpha, x). \tag{3.9}$$

This then gives

$$\det\left(-\partial^2 \delta_{\mu\nu} + (1-1/\xi)\partial_\mu \partial_\nu\right) = \prod \frac{1}{\xi}(k^2(\alpha))^4 = \left(\frac{1}{\xi}\right)^{n'_{\rm L}} \left[\det(-\partial^2)\right]^4, \tag{3.10}$$

where $n'_{\rm L}$ is the number of longitudinal eigenmodes of $-\partial^2\delta_{\mu\nu} + (1-1/\xi)\partial_\mu\partial_\nu$. Now the total number of modes in the one-instanton sector and zero-instanton sector must be equal* since each side of eq. (3.2) is dimensionless, and gauge invariance of eq. (3.2) implies that we should take $n_{\rm L} = n'_{\rm L}$. The zero modes are all transverse, i.e. they satisfy $D_\mu \zeta_\mu = 0$.

Piecing this together eq. (3.2) becomes

$$\frac{\langle 0|0\rangle_I}{\langle 0|0\rangle_0} = C'\frac{1}{g^{4N}}\int \frac{d^{4N}b_i}{\rho^{4N}} e^{-(8\pi^2/g^2 - 4N\ln\rho\mu)} \frac{\det(-\partial^2)}{\det(-D^2)}. \tag{3.11}$$

The quantity $\det(-\partial^2)/\det(-D^2)$ contains ultraviolet divergences so that after renormalization the renormalized quantity will contain an explicit μ-dependence. Since we are only interested in the μ-dependence we can expand $\det(-D^2)$ in powers of the instanton field, keeping only the terms which lead to ultraviolet divergences. Furthermore since we know that in the background field gauge [7] the only divergent terms which can appear in the effective action must be proportional to

$$\int d^4x\, F_{\mu\nu}F_{\mu\nu}\left(= \frac{32\pi^2}{g^2} \quad \text{in the case of an instanton}\right),$$

it is sufficient to expand $\det(-D^2)$ up to second order in the instanton field and look at the coefficient of the ultraviolet divergence which multiplies $\int d^4x\, A^I_\mu \partial^2 A^I_\mu$. This coefficient must then be the coefficient of the ultraviolet divergence which multiplies $-\frac{1}{2}\int d^4x\, F_{\mu\nu}F_{\mu\nu}$.

Now

$$D^2 = \partial^2 + 2igA^I_\mu \partial_\mu - g^2 A^I_\mu A^I_\mu \tag{3.12}$$

* This is strictly only true if the system is placed in a box and an ultraviolet cut-off is imposed on the eigenvalues of the quadratic operator, so that the number of modes is finite.

(where we have used $\partial_\mu A^I_\mu = 0$) and therefore

$$\frac{\det(-\partial^2)}{\det(-D^2)} = \exp\left[-\int d^4x \operatorname{Tr}\ln\left(1 + 2igA^I_\mu \partial_\mu \frac{1}{\partial^2} - g^2 A^I_\mu A^I_\mu \frac{1}{\partial^2}\right)\right], \tag{3.13}$$

where the trace goes over the group indices and for Yang-Mills fields which transform as the adjoint representation under global SU(N) transformations gives us a factor of N. Expanding the integral in the exponent of eq. (3.13) up to quadratic order in the instanton field, A^I_μ gives

$$\int d^4x \operatorname{Tr}\ln\left(1 + 2igA^I_\mu \partial_\mu \frac{1}{\partial^2} - g^2 A^I_\mu A^I_\mu \frac{1}{\partial^2}\right)$$

$$= \int d^4x \operatorname{Tr}\left(2igA_\mu(x)\,\partial_\mu \frac{1}{\partial^2(x,x)} - g^2 A^I_\mu A^I_\mu \frac{1}{\partial^2(x,x)}\right)$$

$$+ 2g^2 \int d^2x\, d^4y \operatorname{Tr}\left(A^I_\mu(x)\,\partial_\mu \frac{1}{\partial^2(x,y)} A^I_\nu(y)\,\partial_\nu \frac{1}{\partial^2(y,x)}\right) + \cdots . \tag{3.14}$$

The integrations over x and y diverge, but after renormalization we obtain a μ-dependent part:

$$-\frac{1}{3}\frac{g^2}{16\pi^2}\int d^4x \ln(-\mu^2/\partial^2)\operatorname{Tr}\left(A^I_\mu(x)\left(\partial^2\delta_{\mu\nu} - \partial_\mu\partial_\nu\right)A^I_\nu(x)\right). \tag{3.15}$$

Using the fact that the only other scale which can accompany the quantity μ is the instanton size, ρ, we have

$$\int d^4x \operatorname{Tr}\ln\left(1 + 2igA^I_\mu \partial_\mu \frac{1}{\partial^2} - g^2 A^I_\mu A^I_\mu \frac{1}{\partial^2}\right)$$

$$= \frac{1}{3}\frac{g^2}{16\pi^2}\ln(\mu\rho)\int d^4x \operatorname{Tr}\left[F_{\mu\nu}(x)F_{\mu\nu}(x)\right] + \text{const.} \tag{3.16}$$

For an instanton background field this is

$$\tfrac{2}{3}N\ln(\mu\rho) + \text{const}, \tag{3.17}$$

so that

$$\det\left(\frac{1}{\partial^2}D^2\right)^{-1} \sim e^{-(2/3)N\ln(\mu\rho)}. \tag{3.18}$$

Inserting this into eq. (3.11) we have finally

$$\frac{\langle 0|0\rangle_I}{\langle 0|0\rangle_0} = C\frac{1}{g^{4N}}\int \frac{\mathrm{d}^{4N}b_i}{\rho^{4N}}\xi^{N/2}\mathrm{e}^{(-8\pi^2/g^2+(11/3)N\ln(\mu\rho))}, \tag{3.19}$$

where C is a μ-independent constant. Note that in the expansion for β defined by eq. (1.12) $\beta_0 = \frac{11}{3}N$ so that the leading-order $\ln\mu$ dependence of $8\pi^2/g^2$ is exactly cancelled by the term $\frac{11}{3}N\ln(\mu\rho)$ in the exponent, as required by the renormalization group.

In the next section we will calculate the two-loop correction to this semi-classical result.

4. $O(g^2)$ corrections to the semi-classical approximation

In going beyond the semi-classical approximation of sect. 3 we consider the whole of the effective action, including the cubic and quartic interactions between the quantum fields as well as the $O(g^2)$ corrections to det J, the jacobian discussed in sect. 2. We encounter four new diagrams shown in figs. 3 and 4. The diagrams of fig. 3 are the usual two-loop "vacuum" graphs where the propagators are the full propagators in the background field A_μ^I, but where the zero modes have been eliminated. There are two divergent (and hence, after renormalisation, μ-dependent) contributions arising from each of these three diagrams. The first is a purely short-distance contribution arising from simultaneous high momenta in both loops

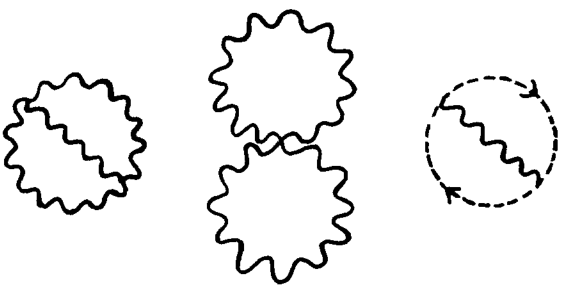

Fig. 3. Two-loop vacuum graphs in the background field A_μ^I (the thick black line indicates a full propagator in the background field).

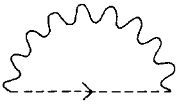

Fig. 4. One-loop diagram arising from the jacobian.

and this serves to renormalise the term $-8\pi^2/g^2$ in the exponent of the semi-classical result. The second is a contribution from the region of phase space where one of the two propagators is long distance, so we only have large momenta in one of the two loops (this latter contribution cannot be studied by expanding in powers of the background field). The diagram of fig. 4 which is encountered at this level due to the bilinear interactions between the Faddeev-Popov ghost fields and the vector fields combines with the "long-distance" contributions from the diagrams of fig. 3 to renormalize the factor $1/g$ for each zero mode in the semi-classical approximation.

We discuss these contributions separately.

4.1. PURELY SHORT-DISTANCE CONTRIBUTION

If the momenta in both loops are large the divergence comes only from the short-distance part of each propagator. In this case the structure of the instanton, and consequently the fact that the zero modes have been eliminated from the propagators, will not be seen and the divergences will be those found for any background field. The calculation can be carried out pertubatively and the diagrams contributing to the divergence are obtained by expanding the propagators in powers of the background field, keeping only those diagrams which contain overlapping divergences. Some examples of these diagrams are shown in fig. 5. The fact that the choice of background field gauge preserves background gauge invariance implies that the counterterm must be proportional to $F_{\mu\nu}F_{\mu\nu}$ so that in order to find the coefficient of the divergent part we need consider only diagrams which are second order in the background field. These diagrams have been evaluated by Abbott [8] and yield the results

$$\frac{17}{3}\left(\frac{g^2}{16\pi^2}\right)^2 N^2 \ln(\mu\rho)\int d^4x\, F_{\mu\nu}F_{\mu\nu} \tag{4.1}$$

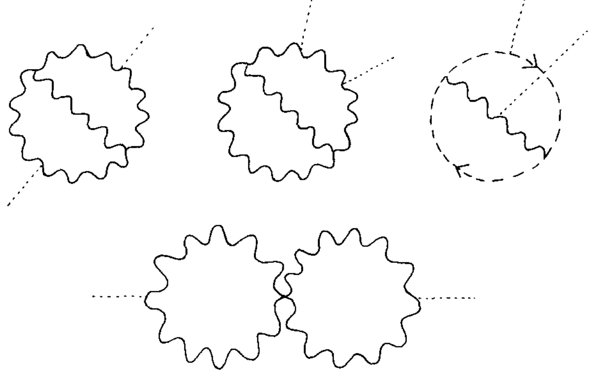

Fig. 5. Examples of Feynman diagrams with only short-distance contributions.

(after renormalisation). This exactly cancels the μ-dependence of the classical part of the effective action $-8\pi^2/g^2$ at the two-loop level.

4.2. LONG-DISTANCE CONTRIBUTIONS

There are also μ-dependent contributions from diagrams in which one of the propagators is taken to be long distance and the other two are taken to be short distance. Examples of these are given in fig. 6. For the long-distance propagator there are enough insertions of the background field so that the loop containing that propagator is ultraviolet convergent. The divergent subgraph in the diagrams where one of the vector propagators is taken to be long distance is the self-energy of the (quantum) vector field. In the presence of a background field the μ-dependent part of the renormalised quantity is given by

$$\Pi^I_{\mu\nu} = \left(\tfrac{13}{3} - \xi\right)\frac{g^2}{16\pi^2} N \ln(\mu\rho)\left\{-D^2\delta_{\mu\nu} + D_\mu D_\nu + 2igF_{\mu\nu}\right\} \tag{4.2}$$

(the quantity in the curly brackets is $O_{\mu\nu}$). This form is guaranteed by the fact that $\Pi^I_{\mu\nu}$ must obey the background field gauge Ward identity

$$D_\mu \Pi^I_{\mu\nu} = 0\,, \tag{4.3}$$

and the coefficient of $O_{\mu\nu}$ must be the same as that obtained in the zero-instanton sector, since the counterterms must be the same in both sectors. Thus the μ-dependent part of the diagrams of the type shown in fig. 6 in which one of the vector propagators is long distance is

$$\left(\tfrac{13}{6} - \tfrac{1}{2}\xi\right)\frac{g^2}{16\pi^2} N \ln(\mu\rho)\int d^4x\, \mathrm{Tr}\left[O_{\mu\nu}\right]\Delta_{\nu\mu}(x,x)\,, \tag{4.4}$$

where $\Delta_{\nu\mu}(x,y)$ is the propagator in the background field, in the absence of zero modes, and a combinatorial factor of $\frac{1}{2}$ has been included. The trace is taken over the group space. Strictly speaking we should only consider the long-distance part of

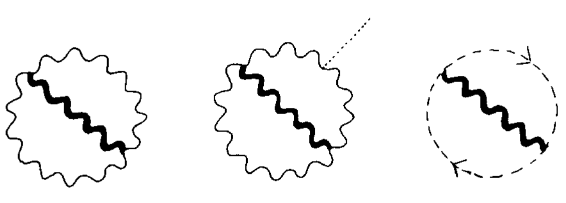

Fig. 6. Examples of Feynman graphs with two short-distance propagators and one long-distance propagator (represented by a thick line to indicate a full propagator in the background field).

the vector propagator $\Delta_{\mu\nu}$. However if we look at the short-distance part by expanding eq. (4.4) to any finite power of the background field we get contributions which exactly cancel the short-distance contributions from similar terms in the zero-instanton sector. There remains however a "non-perturbative" contribution which we now exhibit. We start by rewriting eq. (4.4) as

$$\left(\tfrac{13}{6}-\tfrac{1}{2}\xi\right)\frac{g^2}{16\pi^2}N\ln(\mu\rho)\int \mathrm{d}^4x\left[\mathrm{Tr}\left(O_{\mu\nu}-(1/\xi)D_\mu D_\nu\right)\Delta_{\nu\mu}(x,x)\right.$$
$$\left.+(1/\xi)\mathrm{Tr}\left[D_\mu D_\nu\right]\Delta_{\nu\mu}(x,x)\right]. \qquad (4.5)$$

$\Delta_{\mu\nu}$ is the inverse of $O_{\mu\nu}-(1/\xi)D_\mu D_\nu$, but on to a space from which the zero modes have been eliminated. Consequently

$$\int \mathrm{d}^4x\,\mathrm{Tr}\left(O_{\mu\nu}-(1/\xi)D_\mu D_\nu\right)\Delta_{\nu\mu}(x,x)=n^+, \qquad (4.6)$$

where n^+ is the total number of positive frequency modes. As has been shown in sect. 3, $-D_\mu D_\nu$ projects the n_{L} longitudinal modes from $\Delta_{\mu\nu}$ with eigenvalue ξ:

$$\int \mathrm{d}^4x\,\frac{1}{\xi}\mathrm{Tr}\left[D_\mu D_\nu\right]\Delta_{\nu\mu}(x,x)=-n_{\mathrm{L}}. \qquad (4.7)$$

So the μ-dependence of the diagrams of the type shown in fig. 6 is

$$\left(\tfrac{13}{6}-\tfrac{1}{2}\xi\right)\frac{g^2}{16\pi^2}N\ln(\mu\rho)n_{\mathrm{T}}^+, \qquad (4.8)$$

where $n_{\mathrm{T}}^+(=n^+-n_{\mathrm{L}})$ is the number of transverse positive modes. (Alternatively the result can be derived using the vector propagator of ref. [14].)

In the zero-instanton sector we get a similar contribution with $O_{\mu\nu}$ replaced by $-\partial^2\delta_{\mu\nu}+\partial_\mu\partial_\nu$ and $\Delta_{\mu\nu}$ replaced by the vector propagator in the absence of a background field, namely $(-\delta_{\mu\nu}+(1-\xi)\partial_\mu\partial_\nu)/\partial^2$. This therefore gives rise to the term

$$\left(\tfrac{13}{6}-\tfrac{1}{2}\xi\right)\frac{g^2}{16\pi^2}N\ln(\mu\rho)n'_{\mathrm{T}}, \qquad (4.9)$$

where n'_{T} is the number of transverse modes in the zero-instanton sector. Since $n'_{\mathrm{T}}-n_{\mathrm{T}}^+$ is the number of zero modes $(=4N)$, subtracting the contribution from the zero-instanton sector yields

$$-\left(\tfrac{13}{6}-\tfrac{1}{2}\xi\right)\frac{g^2}{16\pi^2}4N^2\ln(\mu\rho). \tag{4.10}$$

We point out that since the operator $-D^2$ does not have zero modes we do not get a contribution to the ratio $\langle 0|0\rangle_I/\langle 0|0\rangle_0$ when one of the Faddeev-Popov ghost propagators is long distance.

4.3. CONTRIBUTIONS FROM THE CORRECTION TO THE JACOBIAN

The new graph (fig. 4) is a graph which occurs to this order because of the extra interactions between vector fields and Faddeev-Popov ghosts which arise from simultaneous treatment of the gauge-fixing procedure and zero-mode elimination, discussed in sect. 2. The graph contains only one loop, but must be considered at the $O(g^2)$ level because of the $\sqrt{g}$ in the vertex (see fig. 2) and the fact that they replace the factor $1/g$ for each zero mode in the semi-classical approximation. This graph has hitherto been neglected, but as we shall see it provides a μ-dependence which is necessary to provide the expected renormalisation group invariance at the $O(g^2)$ level.

The graph has a divergent (and hence after renormalisation a μ-dependent) contribution when the vector and Faddeev-Popov ghost propagators are both taken with zero insertions of the background field.

From the Feynman rules of fig. 2 this diagram gives

$$g\sum_{i,j=1}^{4N}\bar{c}^i c^j\int \mathrm{d}^4x\,\mathrm{d}^4y\,\zeta_\mu^{j,e}(x)\zeta_\nu^{i,d}(y)f^{abe}f^{cdf}\left(\frac{1}{D^2}\right)^{bd}(x,y)\Delta_{\mu\nu}^{ac}(x,y), \tag{4.11}$$

where i and j run over the discrete zero modes and $c^i, \bar{c}^i$ are the Grassmann variables introduced in sect. 2 for each discrete zero mode. In momentum space eq. (4.11) contains the divergent integral

$$g\sum_{i,j=1}^{4N}\bar{c}^i c^j N\int\frac{\mathrm{d}^4p\,\mathrm{d}^4q}{(2\pi)^8}\zeta_\mu^j(p)\zeta_\nu^i(-p)\frac{1}{(p-q)^2}\left(\frac{\delta_{\mu\nu}}{q^2}-(1-\xi)\frac{q_\mu q_\nu}{q^4}\right), \tag{4.12}$$

which after renormalisation yields the μ-dependence

$$\sum_{i=1}^{4N}N_i^2\bar{c}^i c^i\frac{g}{16\pi^2}\tfrac{1}{2}N\ln(\mu\rho)(3+\xi), \tag{4.13}$$

where we have used the orthogonality of zero modes and N_i is the norm of the ith zero mode. The integration over the Grassmann variables c^i and $\bar{c}^i$ may now be performed. As has been noted in sect. 2 in the semi-classical approximation the integral would yield $-(1/g)N_i^2$ for each zero mode. But to this order we have a correction in which for each zero mode "i" we insert one of the terms in the sum (4.13) instead of the factor $-(1/g)N_i^2$, giving a correction factor of

$$1 - \frac{g^2}{16\pi^2}(3+\xi)\tfrac{1}{2}N\ln(\mu\rho) \tag{4.14}$$

(for each zero mode). Adding these correction factors to (4.10) yields an $O(g^2)$ term

$$-\frac{g^2}{16\pi^2}\frac{44}{3}N^2\ln(\mu\rho). \tag{4.15}$$

This is exactly the term which is required to cancel the μ-dependence of the factor $(1/g)^{4N}$ in the semi-classical result, since

$$\frac{\partial}{\partial\ln\mu}\left(\frac{1}{g(\mu)}\right)^{4N} = \left(\frac{1}{g}\right)^{4N}\left[4N\frac{g^2}{16\pi^2}\beta_0\right], \tag{4.16}$$

with $\beta_0 = \frac{11}{3}N$.

This is the expected result from the fact that the ratio $\langle 0|0\rangle_I/\langle 0|0\rangle_0$ must be renormalization group invariant, but we emphasize that this μ-independence is only achieved when the extra diagrams of fig. 4 are taken into account.

5. The fermion sector

In order to incorporate fermions into the theory we add to the action a term

$$S_F = \int d^4x\left[-i\bar{\psi}\gamma^\mu D_\mu\psi + g\bar{\psi}\gamma^\mu Q_\mu\psi\right]. \tag{5.1}$$

In the semi-classical approximation we keep only the terms which are quadratic in the quantum fields so we drop the second term in the integrand in eq. (5.1). Integration over the Grassmann fields $\bar{\psi}(x)$ and $\psi(x)$ gives a factor [2–5]

$$\det(-i\gamma^\mu D_\mu).$$

However in the case of an instanton background field $-i\gamma^\mu D_\mu$ possesses $2T(R)$ zero modes [2] ($T(R)$ is the eigenvalue of the quadratic Casimir operator of the gauge group in the fermion representation). This factor gives us zero and we conclude that the vacuum-vacuum amplitude in the presence of an instanton vanishes if the fermions are massless. On the other hand there are chiral symmetry breaking operators which have non-zero vacuum expectation values due to instantons, so that the structure of the vacuum is more complicated in the presence of massless fermions, being a superposition of fermion-antifermion condensates.

We choose instead to introduce explicit chiral symmetry breaking by adding a small mass term for the fermions. This mass, m, is chosen to be so small that it raises the eigenvalue of the fermionic zero mode to m, but has a negligible effect on the non-zero fermionic modes*.

Although at first sight it may seem that the interactions of the non-zero modes will therefore not affect the mass m, so it will not receive renormalization in higher order and can be considered to be a μ-independent constant, this picture is inconsistent with the fact that the counterterms that one must add to an action are independent of what (if any) kind of background field one expands about. If we add a mass term to the theory then in the zero-instanton sector that mass does get renormalized and becomes a function of the subtraction point μ. Likewise in the one-instanton sector we expect to see contributions which renormalize the mass parameter. In this section we shall show that there are extra contributions analogous to those discussed in sect. 4 which have a μ-dependence which compensates the μ-dependence of the mass parameter m.

We expand the fermion field in terms of the eigenstates of the operator $-i\gamma^{\mu}D_{\mu}$, with coefficients, a, which are Grassmann variables;

$$\psi(x)=\sum_{i=1}^{2T(R)} a_0^i u_0^i+\int \mathrm{d}\alpha\left(a^{\mathrm{L}}(\alpha)u^{\mathrm{L}}(\alpha,x)+a^{\mathrm{R}}(\alpha)u^{\mathrm{R}}(\alpha,x)\right), \tag{5.2}$$

where the indices L and R refer to left and right chiralities respectively $u^{\mathrm{L(R)}}(\alpha,x)$ are the non-zero modes,

$$-i\gamma^{\mu}D_{\mu}u^{\mathrm{L(R)}}(\alpha,x)=\lambda(\alpha)u^{\mathrm{R(L)}}(\alpha,x), \tag{5.3}$$

and $u_0^i(x)$ are the zero modes (which are always right-handed)

$$-i\gamma^{\mu}D_{\mu}u_0^i(x)=0. \tag{5.4}$$

Assuming that all these modes are normalized to 1 the fermionic part of the action

* This can be achieved if the system is put in a box of dimension R. The smallest non-zero eigenvalue will be $\sim 1/R$, so we require $mR \ll 1$.

may now be written

$$S_F = -im \sum_{i=1}^{2T(R)} a_0^{i*} a_0^i + \int \mathrm{d}\alpha \left[\lambda(\alpha)\left(a^L(\alpha)^* a^L(\alpha) + a^R(\alpha)^* a^R(\alpha)\right) \right.$$
$$\left. - im\left(a^L(\alpha)^* a^R(\alpha) + \text{h.c.}\right) \right]$$
$$+ g \int \mathrm{d}^4x\, \mathrm{d}\alpha\, \mathrm{d}\beta \left(a^L(\alpha)^* a^L(\beta) \bar{u}^L(\alpha, x) \gamma^\mu Q_\mu u^L(\beta, x) + \mathrm{L} \to \mathrm{R} \right)$$
$$+ g \sum_{i,j} \int \mathrm{d}^4x\, a_0^{i*} a_0^j \bar{u}_0^i(x) \gamma^\mu Q_\mu u_0^j(x)$$
$$+ g \sum_i \int \mathrm{d}^4x\, \mathrm{d}\alpha \left[a^R(\alpha)^* a_0^i \bar{u}^R(\alpha, x) \gamma^\mu Q_\mu u_0^i(x) \right.$$
$$\left. + a_0^{i*} a^L(\alpha) \bar{u}_0^i(x) \gamma^\mu Q_\mu u^L(\alpha, x) \right]. \tag{5.5}$$

In the semi-classical approximation we drop the terms which are cubic in the quantum fields (and neglect im with respect to $\lambda(\alpha)$) so that the ratio $\langle 0|0\rangle_I / \langle 0|0\rangle_0$ acquires a factor

$$\frac{\displaystyle \int \prod_i \mathrm{d}a_0^i \mathrm{d}a_0^{i*} \prod_\alpha \mathrm{d}a^L(\alpha) \mathrm{d}a^L(\alpha)^* \mathrm{d}a^R(\alpha) \mathrm{d}a^R(\alpha)^* \times \exp\left[-im \sum_{i=1}^{2T(R)} a_0^{i*} a_0^i + \int \mathrm{d}\alpha \lambda(\alpha)\left(a^L(\alpha)^* a^L(\alpha) + \mathrm{L} \to \mathrm{R}\right) \right]}{\displaystyle \int D[\bar{\psi}] D[\psi] \exp\left[-i \int \mathrm{d}^4x\, \bar{\psi}(x) \gamma^\mu \partial_\mu \psi(x) \right]}, \tag{5.6}$$

where in the denominator we have dropped the mass m, since the operator $-i\gamma^\mu \partial_\mu$ has no zero modes.

The integral over the (Grassmann) coefficients of the zero modes gives rise to a factor

$$\left(\frac{m}{\mu} \right)^{2T(R)},$$

where the $\mu^{2T(R)}$ in the denominator is required on dimensional grounds and can be explained in terms of Pauli-Villars regularization (see refs. [2, 6]). It has been shown [12, 13] that for each non-zero mode of $-i\gamma^\mu D_\mu$ with eigenvalue $\lambda(\alpha)$ there corresponds a mode of $-D^2$ with eigenvalue $\lambda^2(\alpha)$, and conversely for every eigenmode,

$\Omega(\alpha, x)$ of $-D^2$ with eigenvalue $\lambda^2(\alpha)$ there corresponds a pair of spinors $u^{L}(\alpha, x)$, $u^{R}(\alpha, x)$ given by

$$u^{L}(\alpha, x) = \tfrac{1}{2}(1-\gamma^5)\Omega(\alpha, x)v, \tag{5.7a}$$

where v is a constant spinor

$$u^{R}(\alpha, x) = -i\gamma^{\mu}D_{\mu}\Omega(\alpha, x)\tfrac{1}{2}(1+\gamma^5)v, \tag{5.7b}$$

which both obey eq. (5.3).

Similarly in the zero-instanton sector to each eigenmode of $-i\gamma^{\mu}\partial_{\mu}$ with eigenvalue $k(\alpha)$ there corresponds an eigenmode of $-\partial^2$ with eigenvalue $k^2(\alpha)$ and conversely for each eigenmode of $-\partial^2$ with eigenvalue $k^2(\alpha)$ there exists a pair of spinors $\chi^{L,R}$ such that

$$-i\gamma^{\mu}\partial_{\mu}\chi^{L(R)} = k(\alpha)\chi^{R(L)}. \tag{5.8}$$

The expression (5.6) therefore becomes (up to a constant)

$$\left(\frac{m}{\mu}\right)^{2T(R)} \frac{\det(-D^2)}{\det(-\partial^2)}. \tag{5.9}$$

Following the steps discussed in sect. 3 the μ-dependent part of this is

$$\left(\frac{m}{\mu}\right)^{2T(R)} \exp\left[\frac{1}{3}\frac{g^2}{16\pi^2}\ln(\mu\rho)\int d^4x\, \mathrm{Tr}(F_{\mu\nu}F_{\mu\nu})\right], \tag{5.10}$$

but this time the trace refers to the trace of the generators in the representation of the fermions, which gives us a factor of $T(R)$. Using

$$\mathrm{Tr}\int d^4x\, F_{\mu\nu}F_{\mu\nu} = \frac{32}{\pi^2}T(R), \tag{5.11}$$

and multiplying this factor by the semi-classical result in the pure Yang-Mills sector, eq. (3.18), we obtain

$$\frac{\langle 0|0\rangle_I}{\langle 0|0\rangle_0} \sim \frac{1}{g^{4N}}\int \frac{d^{4N}b_i}{\rho^{4N}}(m\rho)^{2T(R)}$$
$$\times \exp\left[-\frac{8\pi^2}{g^2} + \left(\tfrac{11}{3}N - \tfrac{4}{3}T(R)\right)\ln(\mu\rho)\right] + O(g^2). \tag{5.12}$$

Note that the extra $\ln\mu$ dependence from the fermion sector is exactly that required to cancel the fermion contribution to the μ-dependence of the term $-8\pi^2/g^2$ in the exponent.

In order to obtain the $O(g^2)$ correction to the fermionic contribution we must consider the whole of S_F (eq. (5.5)). Since we are integrating over Grassmann variables we keep only the terms linear in each Grassmann variable. Therefore, relative to the semi-classical approximation, for each Grassmann variable which is obtained from a cubic term in eq. (5.5) in the expansion of $\exp(-S_F)$, we must divide by the corresponding coefficient in the quadratic part (i.e. we have a factor $1/\lambda(\alpha)$ when we consider a term proportional to $g^2 a^L(\alpha)^* a^L(\alpha)$, and $1/m$ when we consider a term proportional to $g^2 a_0^{i*} a_0^i$ etc.) This gives a correction factor which we write

$$1 + g^2(A + B + C), \tag{5.13}$$

where the contribution A is that from the non-zero modes alone, contribution B comes from the interaction between the fermionic zero modes and the vector field Q_μ, and finally contributions C comes from the interaction between the fermionic zero modes, the fermionic non-zero modes and Q_μ.

Thus

$$A = -\int \frac{d^4x\, d^4y\, d\alpha\, d\beta}{\lambda(\alpha)\lambda(\beta)} \Big(\bar{u}^L(a, x) \gamma^\mu \tau^a u^L(\beta, x) \bar{u}^L(\beta, y) \times \gamma^\nu \tau^b u^L(\alpha, y) \Delta^{ba}_{\nu\mu}(y, x) + L \to R \Big), \tag{5.14}$$

where $\Delta^{ab}_{\mu\nu}(x, y)$ is the propagator of the quantum vector field (a, b are group indices and τ^a, τ^b are the group generators in the fermion representation):

$$B = -\int \frac{d^4x\, d^4y}{m^2} \sum_{i,j=1}^{2T(R)} \bar{u}_0^i(x) \gamma^\mu \tau^a u_0^j(x) \bar{u}_0^j(y) \gamma^\nu \tau^b u_0^i(y) \Delta^{ba}_{\nu\mu}(y, x), \tag{5.15}$$

$$C = \frac{m}{m} \sum_{i=1}^{2T(R)} \int \frac{d^4x\, d^4y\, d\alpha}{\lambda^2(\alpha)} \bar{u}^R(\alpha, x) \gamma^\mu \tau^a u_0^i(x) \bar{u}_0^i(y) \gamma^\nu \tau^b u^L(\alpha, y) \Delta^{ba}_{\nu\mu}(y, x), \tag{5.16}$$

where in C we needed one power of $-ima^{L*}(\alpha) a^R(\alpha)$ in order to have one and only one power of each Grassmann variable in the integrand. The minus signs in front of A and B arise because it was necessary to interchange the order of an odd number of pairs of Grassmann variables (this is equivalent to the minus sign required for each fermion loop).

We consider these contributions individually.

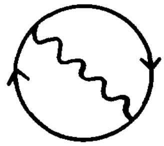

Fig. 7. Two-loop vacuum graph, with fermions, in the background field A^I_μ (represented by thick lines to indicate full propagators).

5.1. CONTRIBUTION A

$$\int \mathrm{d}\alpha \frac{u^{\mathrm{L}}(\alpha, x)\bar{u}^{\mathrm{L}}(\alpha, y)}{\lambda(\alpha)}$$

is the propagator for the left-handed fermions, $\Delta^{\mathrm{L}}_{\mathrm{F}}(x, y)$ (the same is true for L replaced by R, but in this case the propagator is understood to be the propagator of the non-zero modes only). Therefore we may write A as

$$A = -\int \mathrm{d}^4x\,\mathrm{d}^4y\left[\mathrm{Tr}\left(\Delta^{\mathrm{L}}_{\mathrm{F}}(x, y)\gamma^\mu\tau^a\Delta^{\mathrm{L}}_{\mathrm{F}}(y, x)\gamma^\nu\tau^b\right) + \mathrm{L} \to \mathrm{R}\right], \tag{5.17}$$

where the trace is over both Dirac matrices and the group generators. This is the usual two-loop diagram shown in fig. 7, where the propagators are taken to represent propagation in the instanton background field. As in sect. 5 in the case of vector particles alone the μ-dependent part of this diagram has two contributions:

5.1.1. Short-distance contributions. These are the diagrams in which all the propagators are considered to be short distance. They are obtained by expanding the propagators in a power series in the instanton field, keeping only the diagrams which contain overlapping divergences. The diagrams are shown in fig. 8. As has been discussed in sect. 4 since these are short-distance contributions and the zero modes are long-distance effects, these contributions are the same as those obtained by performing two-loop calculations in an ordinary background field. As has been shown by Abbott this contribution must be the fermionic contribution to

$$\left(\frac{g^2}{16\pi^2}\right)^2 2\beta_1 \ln(\mu\rho)\int \mathrm{d}^4x\,\tfrac{1}{4}F_{\mu\nu}F_{\mu\nu},$$

where β_1 is defined in eq. (1.12).

Using the known fermion contribution to β_1, the short-distance contribution to A is

$$A^{(a)} = -\frac{1}{16\pi^2}\ln(\mu\rho)\left(\tfrac{20}{3}NT(R) + 4C_{\mathrm{F}}T(R)\right), \tag{5.18}$$

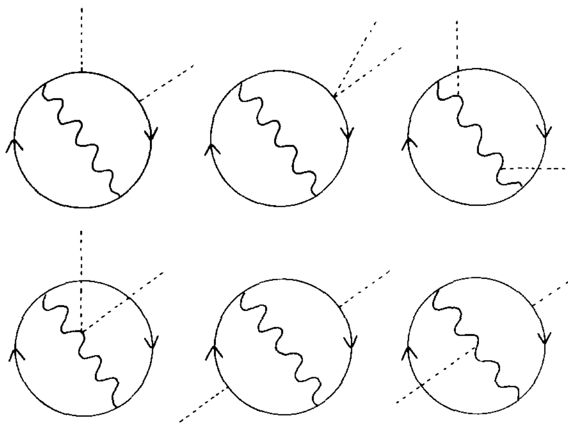

Fig. 8. The diagrams yielding the purely short-distance fermionic contribution.

with C_F defined by

$$\tau^a_{ij}\tau^a_{jk} = C_F \delta_{ik}, \tag{5.19}$$

τ^a_{ij} being the generators of the gauge group in the fermion representation.

5.1.2. Long-distance contribution. As has also been discussed in sect. 4 there are μ-dependent contributions arising from diagrams in which one of the propagators is long distance and the other two are short distance giving rise to a divergent (and hence after renormalization a μ-dependent) subgraph. For the case where the long-distance propagator is the vector propagator (see fig. 9), the divergent subgraphs are the fermionic contribution to the self-energy of the vector field and we get a contribution to A:

$$A_1^{(b)} = -\frac{1}{4\pi^2}\frac{1}{3}T(R)\ln(\mu\rho)\int d^4x\, \mathrm{Tr}\big(O_{\mu\nu} - (1/\xi)D_\mu D_\nu\big)\Delta_{\nu\mu}(x,x). \tag{5.20}$$

As has been discussed in sect. 4 this compensates the fermionic contribution to the μ-dependence of the factor $(1/g)^{4N}$ in the semi-classical approximation. For the case where one of the fermion propagators is the long-distance propagator (see figs. 10),

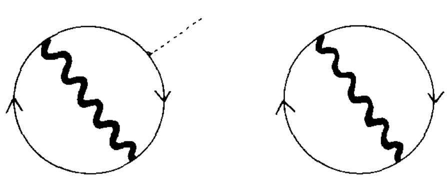

Fig. 9. Fermionic contribution with long-distance vector propagator.

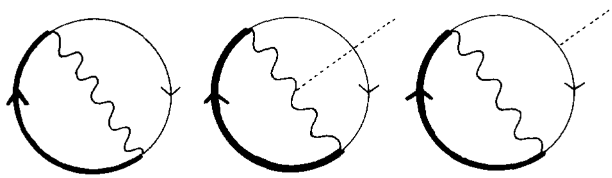

Fig. 10. Fermionic contribution with long-distance fermion propagator.

the divergent subgraph is the self-energy of the fermion which has a μ-dependence

$$-\frac{g^2}{16\pi^2}2C_{\mathrm{F}}\xi \ln\mu(-i\gamma^{\nu}D_{\nu}).$$

The contribution to A is therefore

$$A_2^{(b)}=\frac{1}{8\pi^2}C_{\mathrm{F}}\xi\ln(\mu\rho)\int \mathrm{d}^4x\,\mathrm{Tr}\big[-i\gamma^{\nu}D_{\nu}(\Delta_{\mathrm{F}}^{\mathrm{L}}(x,x)+\Delta_{\mathrm{F}}^{\mathrm{R}}(x,x)\big]. \quad (5.21)$$

From the zero-instanton sector we get a similar contribution:

$$\frac{1}{8\pi^2}C_{\mathrm{F}}\xi\ln(\mu\rho)\int' \mathrm{d}^4x\,\mathrm{Tr}\big(-i\gamma^{\nu}\partial_{\nu}\Delta_{\mathrm{F}}^{0}(x,x)\big), \quad (5.22)$$

where Δ_{F}^{0} is the fermion propagator in the zero-instanton sector. The difference between the expression in eqs. (5.22) and (5.21) comes only from the absence of the $2T(R)$ fermion zero modes from $\Delta_{\mathrm{F}}^{\mathrm{R}}$. Thus we are left with a contribution to A:

$$A_2^{(b)}=-\frac{1}{4\pi^2}\xi C_{\mathrm{F}}T(R)\ln(\mu\rho). \quad (5.23)$$

5.2. CONTRIBUTION B

This is a tree diagram consisting of a vector particle propagating between two currents given by $\bar{u}_0^i(x)\gamma^{\mu}u_0^j(x)$. This cannot contribute a μ-dependence and gives rise only to a constant correction of order g^2. We shall not consider it further.

5.3. CONTRIBUTION C

Using the expression given in eq. (5.7) for $u^{\mathrm{L(R)}}(\alpha,x)$ this may be written

$$C=-\sum_{i=1}^{2T(R)}\int \mathrm{d}^4x\,\mathrm{d}^4y\,\mathrm{d}\alpha\,\Omega(\alpha,x)\frac{1}{\lambda(\alpha)^2}\Omega(\alpha,y)\bar{v}\gamma^{\mu}\tau^{a}u_0^i(x)\bar{u}_0^i(y)\gamma^{\nu}\tau^{b}v\Delta_{\nu\mu}^{ba}(y,x). \quad (5.24)$$

Now

$$\int \mathrm{d}\alpha\, \Omega(\alpha, x) \frac{1}{\lambda(\alpha)^2} \Omega(\alpha, y) = \left(\frac{1}{D^2}\right)_{x,y}. \tag{5.25}$$

The above expression is divergent (and hence μ-dependent after renormalization) when $1/D^2$ and the vector propagator $\Delta^{ab}_{\mu\nu}(y, x)$ are taken with zero-instanton insertions.

The divergent part of C is given by

$$C = \sum_{i=1}^{2T(R)} \int \mathrm{d}^4x\, \mathrm{d}^4y\, \bar{v}\gamma^\mu \tau^a u_0^i(x) \bar{u}_0^i(x) \gamma^\nu \tau^b v \delta_{ab}$$
$$\times \left(\frac{1}{\partial^2(x, y)} \left(\delta_{\mu\nu} - (1-\xi) \frac{\partial_\mu \partial_\nu}{\partial^2(y, x)} \right) \frac{1}{\partial^2(y, x)} \right). \tag{5.26}$$

This is represented diagrammatically in fig. 11. After renormalization the μ-dependent part is given by

$$C = \sum_{i=1}^{2T(R)} \frac{3+\xi}{32\pi^2} \ln(\mu\rho) \int \mathrm{d}^4x\, \mathrm{d}^4y\, \bar{v}\gamma^\mu \tau^a u_0^i(x) \bar{u}_0^i(x) \gamma^\nu \tau^b v \delta_{ab} \delta_{\mu\nu} \delta^4(x-y). \tag{5.27}$$

Using the normalization of the fermionic zero modes

$$\int \mathrm{d}^4x\, \bar{u}_0^i(x) u_0^i(x) = I, \tag{5.28}$$

where I is taken to be the unit matrix in both Dirac space and the group space, and $\bar{v}v = 1$, eq. (5.27) becomes

$$C = \sum_{i=1}^{2T(R)} \frac{C_\mathrm{F}}{8\pi^2}(3+\xi)\ln(\mu\rho) = \frac{C_\mathrm{F} T(R)}{4\pi^2}(3+\xi)\ln(\mu\rho). \tag{5.29}$$

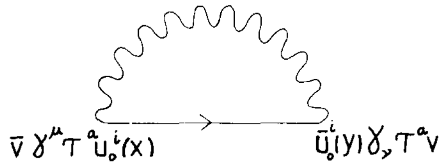

Fig. 11. Fermionic contribution arising as a result of fermion zero modes.

Adding together C and $A_2^{(b)}$ (eq. (5.23)) we see that the gauge-dependent term proportional to ξ cancels and we are left with a correction factor

$$1+\frac{3g^2}{4\pi^2}C_{\mathrm{F}}T(R)\ln(\mu\rho), \tag{5.30}$$

which compensates the μ-dependence of the factor $m^{2T(R)}$ in the semi-classical approximation eq. (5.12).

Thus we see that although the fermion mass is chosen to be small compared with the lowest eigenvalue of the non-zero fermion modes so that the mass does not acquire the expected correction from the usual higher-order terms (contributions A), this correction does arise from loop diagrams involving the interactions between fermionic zero modes, the fermionic non-zero modes and the vector field.

The remaining contributions A^a and $A_1^{(b)}$ provide the fermionic contributions to the correction terms

$$\frac{g^2}{16\pi^2}(\beta_1-4N\beta_0)\ln(\mu\rho),$$

which have been discussed in sect. 4.

6. Summary and conclusions

We start this section with a brief summary of our results. For an SU(N) gauge theory the semi-classical calculation plus the renormalisation group imply that the vacuum energy in the presence of an instanton background field is the integral over collective coordinates of the following expression:

$$m(\mu)^{2T(R)}\frac{1}{g^{4N}(\mu)}\exp\left[-\frac{8\pi^2}{g^2(\mu)}+\beta_0\ln(\mu\rho)+\frac{g^2(\mu)}{16\pi^2}\right.$$
$$\left.\times(\beta_1-4N\beta_0-2T(R)\gamma_m)\ln(\mu\rho)+\cdots\right], \tag{6.1}$$

where γ_m is the anomalous dimension of the fermion mass. We have explicitly verified this expression as an instructive exercise in the calculation of higher-order quantum corrections in the presence of a background field with zero modes. We have found that the O(g^2) contribution has a number of sources. The term

$$\frac{g^2}{16\pi^2}\beta_1\ln(\mu\rho) \tag{6.2}$$

in the exponent comes entirely from the short-distance region of phase space and is therefore independent of the background field. It can be evaluated using standard perturbative methods from the diagrams of fig. 3 and fig. 7. These diagrams have additional contributions in which long-distance effects in one propagator have to be taken into account. These contributions cannot be evaluated by expanding the propagator and keeping only a finite number of interactions with the background field, such an expansion always cancels against similar contributions in the zero-instanton sector, one has instead to keep the full expression for the propagator. These "long-distance" contributions, which do not cancel by similar ones in the zero-instanton sector, are proportional to the number of zero modes and are given by

$$\frac{g^2}{16\pi^2}(2N\gamma_Q)\ln(\mu\rho) - \frac{g^2}{16\pi^2}(2T_R\gamma_F)\ln(\mu\rho), \tag{6.3}$$

where γ_Q and γ_F are the anomalous dimensions of the quantum gauge field and fermion field respectively. The first term in (6.3) corresponds to long-distance effects from a gauge field propagator and the second from long-distance effects from a fermion propagator. Next we had to calculate the corrections to the jacobian corresponding to the change of variables from the gauge zero modes to the collective coordinates. The evaluation of these corrections involved introducing new Grassmann integration variables and new terms into the effective action (see sect. 2). These corrections to the jacobian gave a contribution

$$\frac{g^2}{16\pi^2}[-2N\gamma_Q - 4N\beta_0]\ln(\mu\rho). \tag{6.4}$$

Finally we had to treat the fermionic zero modes. We eliminated these by introducing a mass term for the fermions. Although this mass m was assumed to be small, since in the semi-classical approximation we obtain one factor of m for each fermionic zero mode, we had to keep the corresponding terms proportional to m in higher-order corrections also. Thus we found an additional contribution:

$$-\frac{g^2}{16\pi^2}2T_R(\gamma_m - \gamma_F)\ln(\mu\rho). \tag{6.5}$$

Summing up (6.2) to (6.5) we find the correct two-loop contribution (6.1).

Although we have chosen to work in a single instanton background field, most of the novel features of this calculation are more general. In particular the treatment of zero modes presented in sects. 2 and 4 will apply to all background fields which are solutions of the equations of motion and which break some of the symmetries of the theory.

Beyond two loops the delicate question of renormalisation scheme dependence will arise. This is particularly relevant in the supersymmetric case discussed in the introduction, where the formal arguments of ref. [6] were based on the assumption that supersymmetric regularisation and renormalisation were used. Even in the two-loop case discussed in this paper our different treatment of the bosonic and fermionic zero modes clearly would break supersymmetry. Thus to make contact with ref. [6] we have to introduce grassmanian collective coordinates for the fermionic zero modes and calculate the corresponding higher-order corrections. The results of such a study will be presented elsewhere.

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